Comparing Unlike Fractions in Real Situations
Answer Key

Grade 4–5 Mathematics

Quick Answers

1. —
2. $\frac{3}{4}$
3. —
4. —
5. $\frac{7}{10}$
6. —
7. $\frac{1}{2}$, $\frac{3}{5}$, $\frac{7}{10}$
8. —
9. —
10. $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{12}$

Curriculum

Level: Grade 4–5 Mathematics
4.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 10 tagged checks.

1. Ana $\frac{2}{5}$ of a bottle vs. Ben $\frac{3}{4}$ of the same bottle. Half of five fifths is $\frac{2.5}{5}$, so $\frac{2}{5}$ is *less* than $\frac{1}{2}$. Half of four fourths is $\frac{2}{4}$, and $\frac{3}{4}$ is more than that, so $\frac{3}{4}$ is *greater* than $\frac{1}{2}$. One fraction sits below the benchmark and the other above it, so no common denominator is needed. Ben has more water. $\boxed{\frac{2}{5} < \frac{3}{4}}$

2. $\frac{3}{4}$ of a tray written in twelfths. The tray is cut into 12 equal pieces, so the new denominator is 12. Since $4 \times 3 = 12$, multiply the numerator by the same factor: $\frac{3}{4} = \frac{3\times3}{4\times3}$. Dev took 9 of the 12 pieces. $\boxed{\frac{9}{12}}$

Why multiplying top and bottom is legal: $\frac{3}{3} = 1$, and multiplying by 1 cuts each piece into smaller pieces without changing how much cornbread is on the plate.

3. Sam $\frac{2}{3}$ vs. Kim $\frac{5}{8}$. Both are just above $\frac{1}{2}$, so the benchmark cannot separate them — use the common denominator 24 ($3 \times 8 = 24$).

$$\frac{2}{3} = \frac{2\times8}{3\times8} = \frac{16}{24}, \qquad \frac{5}{8} = \frac{5\times3}{8\times3} = \frac{15}{24}$$

With the same size pieces, 16 pieces beat 15 pieces. Sam walked farther. $\boxed{\frac{2}{3} > \frac{5}{8}}$

4. Morning crew $\frac{4}{9}$ vs. afternoon crew $\frac{5}{6}$. Half of nine ninths is $\frac{4.5}{9}$, so $\frac{4}{9}$ falls just short of $\frac{1}{2}$. Half of six sixths is $\frac{3}{6}$, and $\frac{5}{6}$ is well past it. Below the benchmark is always less than above it. $\boxed{\dfrac{4}{9} < \dfrac{5}{6}}$

5. $\frac{7}{10}$ of a cup written in fortieths. The target denominator is 40 and $10 \times 4 = 40$, so multiply the numerator by 4 as well: $\frac{7}{10} = \frac{7\times4}{10\times4}$. $\boxed{\dfrac{28}{40}}$

6. Mia $\frac{5}{6}$ vs. Jonah $\frac{7}{9}$. Both are above $\frac{1}{2}$, so build a common denominator. The least common multiple of 6 and 9 is 18.

$$\frac{5}{6} = \frac{5\times3}{6\times3} = \frac{15}{18}, \qquad \frac{7}{9} = \frac{7\times2}{9\times2} = \frac{14}{18}$$

15 eighteenths is more than 14 eighteenths, so Mia has read more. $\boxed{\dfrac{5}{6} > \dfrac{7}{9}}$

7. Order $\frac{3}{5}$, $\frac{1}{2}$, $\frac{7}{10}$ from least to greatest. Every denominator divides 10, so tenths is the common unit: $\frac{3}{5} = \frac{6}{10}$, $\frac{1}{2} = \frac{5}{10}$, and $\frac{7}{10}$ already has it. Order the numerators $5 < 6 < 7$, then rewrite the fractions in their original form. The Baker family planted least, the Chen family most. $\boxed{\dfrac{1}{2} < \dfrac{3}{5} < \dfrac{7}{10}}$

8. $\frac{7}{8}$ full vs. $\frac{9}{10}$ full. Both are above $\frac{1}{2}$, so measure the missing part instead: the first tank is $\frac{1}{8}$ from full, the second is $\frac{1}{10}$ from full. With the same numerator, the fraction with the *larger* denominator is the smaller amount, so $\frac{1}{10} < \frac{1}{8}$: the second tank is missing less. Missing less means holding more. $\boxed{\dfrac{7}{8} < \dfrac{9}{10}}$

Check with a common denominator: $\frac{7}{8} = \frac{35}{40}$ and $\frac{9}{10} = \frac{36}{40}$, and $35 < 36$. Same conclusion.

9. Recipe A $\frac{11}{15}$ vs. Recipe B $\frac{3}{4}$. Both sit between $\frac{1}{2}$ and 1, so the benchmark is useless here. The least common multiple of 15 and 4 is 60.

$$\frac{11}{15} = \frac{11\times4}{15\times4} = \frac{44}{60}, \qquad \frac{3}{4} = \frac{3\times15}{4\times15} = \frac{45}{60}$$

The recipes differ by only $\frac{1}{60}$ of a cup, and Recipe B is the larger. $\boxed{\dfrac{11}{15} < \dfrac{3}{4}}$

10. Order $\frac{7}{12}$, $\frac{5}{8}$, $\frac{2}{3}$, $\frac{3}{4}$ from greatest to least. All four denominators divide 24, so 24 is the common unit.

$$\frac{7}{12} = \frac{14}{24}, \qquad \frac{5}{8} = \frac{15}{24}, \qquad \frac{2}{3} = \frac{16}{24}, \qquad \frac{3}{4} = \frac{18}{24}$$

Ordering the numerators from greatest to least gives $18 > 16 > 15 > 14$, so Nia walked the farthest and Rosa the least. $\boxed{\frac{3}{4} > \frac{2}{3} > \frac{5}{8} > \frac{7}{12}}$

Grading note: accept any correct common denominator (a student who uses 48 or 72 gets the same order). The order is what is being assessed.